

Дмитрий Коваленко

задние 20

1) мейк

```
obj-m += test1.o
#ccflags-y += -g -DDEBUG

all:
    make -C /lib/modules/7.0.0-14-generic/build M=$(PWD) modules

clean:
    make -C /lib/modules/7.0.0-14-generic/build M=$(PWD) clean
```

2) test1.c

```
#include <linux/module.h>
#include <linux/kernel.h>

int init_module(void)
{
    pr_info("My test module loaded!!!!\n");

    return 0;
}

void cleanup_module(void)
{
    pr_info("My test module unloaded!!!!\n");
}

MODULE_LICENSE("GPL");
```

3) make и применим

```
komal@komal-VirtualBox:~/Desktop/task_20$ uname -r
7.0.0-14-generic
komal@komal-VirtualBox:~/Desktop/task_20$ make
make -C /lib/modules/7.0.0-14-generic/build M=/home/komal/Desktop/task_20 modules
make[1]: Entering directory '/usr/src/linux-headers-7.0.0-14-generic'
make[2]: Entering directory '/home/komal/Desktop/task_20'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
You are using: gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
warning: pahole version differs from the one used to build the kernel
The kernel was built with: 131
You are using: 0
CC [M] test1.o
MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in test1.o
CC [M] test1.mod.o
CC [M] .module-common.o
LD [M] test1.ko
BTF [M] test1.ko
Skipping BTF generation for test1.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/komal/Desktop/task_20'
make[1]: Leaving directory '/usr/src/linux-headers-7.0.0-14-generic'
komal@komal-VirtualBox:~/Desktop/task_20$ sudo ins
insmod          install-info      install-sgmlcatalog
install         installkernel    instmodsh
komal@komal-VirtualBox:~/Desktop/task_20$ sudo insmod test1.ko
[sudo: authenticate] Password:
komal@komal-VirtualBox:~/Desktop/task_20$
```

4) посмотрим логи

```
komal@komal-VirtualBox:~/Desktop/task_20$ sudo insmod test1.ko
[sudo: authenticate] Password:
komal@komal-VirtualBox:~/Desktop/task_20$ lsmod
Module                Size  Used by
test1                  12288  0
snd_seq_dummy          12288  0
snd_hrtimer            12288  1
komal@komal-VirtualBox:~/Desktop/task_20$ sudo dmesg | tail -n3
[ 1795.955242] test1: loading out-of-tree module taints kernel.
[ 1795.955247] test1: module verification failed: signature and/or required key missing - tainting kernel
[ 1795.955647] My test module loaded!!!!
komal@komal-VirtualBox:~/Desktop/task_20$ sudo rmmod test1
komal@komal-VirtualBox:~/Desktop/task_20$ sudo dmesg | tail -n3
[ 1795.955247] test1: module verification failed: signature and/or required key missing - tainting kernel
[ 1795.955647] My test module loaded!!!!
[ 1950.786660] My test module unloaded!!!!
komal@komal-VirtualBox:~/Desktop/task_20$
```

5) создадим test3.c

```
komal@komal-VirtualBox:~/Desktop/task_20$ make
make -C /lib/modules/7.0.0-14-generic/build M=/home/komal/Desktop/task_20 modules
make[1]: Entering directory '/usr/src/linux-headers-7.0.0-14-generic'
make[2]: Entering directory '/home/komal/Desktop/task_20'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
You are using:          gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
warning: pahole version differs from the one used to build the kernel
The kernel was built with: 131
You are using:          0
CC [M]  test3.o
MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in test3.o
CC [M]  test3.mod.o
CC [M]  .module-common.o
LD [M]  test3.ko
BTF [M] test3.ko
Skipping BTF generation for test3.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/komal/Desktop/task_20'
make[1]: Leaving directory '/usr/src/linux-headers-7.0.0-14-generic'
komal@komal-VirtualBox:~/Desktop/task_20$ sudo ins
insmod          install-info      install-sgmlcatalog
install         installkernel  instmodsh
komal@komal-VirtualBox:~/Desktop/task_20$ sudo insmod test3.ko
komal@komal-VirtualBox:~/Desktop/task_20$ sudo dmesg | tail -n3
[ 1795.955647] My test module loaded!!!!
[ 1950.786660] My test module unloaded!!!!
[ 2056.677855] Module loaded! Major number: 239
```

6) создаем test3 и выводим

```
komal@komal-VirtualBox:~/Desktop/task_20$ sudo mknod test3 c 239 0
komal@komal-VirtualBox:~/Desktop/task_20$ ls -la test3
crw-r--r-- 1 root root 239, 0 Jun 18 22:05 test3
komal@komal-VirtualBox:~/Desktop/task_20$ sudo chmod 666 test
test1.c      test3.c      test3.mod    test3.mod.o
test3        test3.ko     test3.mod.c  test3.o
komal@komal-VirtualBox:~/Desktop/task_20$ sudo chmod 666 test3
komal@komal-VirtualBox:~/Desktop/task_20$ ls -la test3
crw-rw-rw- 1 root root 239, 0 Jun 18 22:05 test3
komal@komal-VirtualBox:~/Desktop/task_20$ cat test3
Hello!

komal@komal-VirtualBox:~/Desktop/task_20$ echo "ffdimaff" > test3
komal@komal-VirtualBox:~/Desktop/task_20$ cat test3
ffdimaff
komal@komal-VirtualBox:~/Desktop/task_20$
```